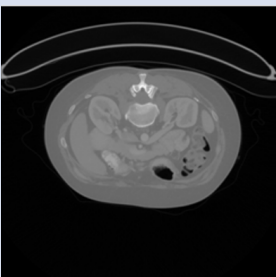


Data Preparation

DCM File



(512x512)

Image

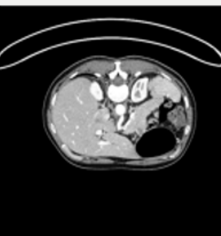
- > HU: [50,350]
- > Pixel normalize

Mask

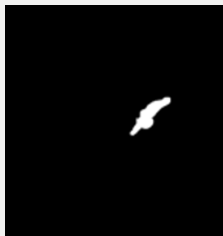
- > Threshold: pixel >= 10 = 1
 : pixel < 10 = 0
- > Bounding box

Only cases where the size of the pancreas 4% or more of the entire image were extracted.

Result (image, mask, box coordinates)



Original



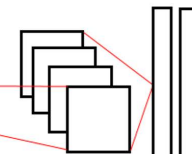
Mask

Box coordinates:
(105,127), (238,238)
(xmin, ymin, xmax, ymax)

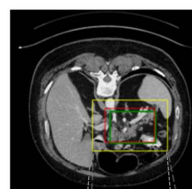
Segmentation

1-Stage

Faster R-CNN (based ResNet50)

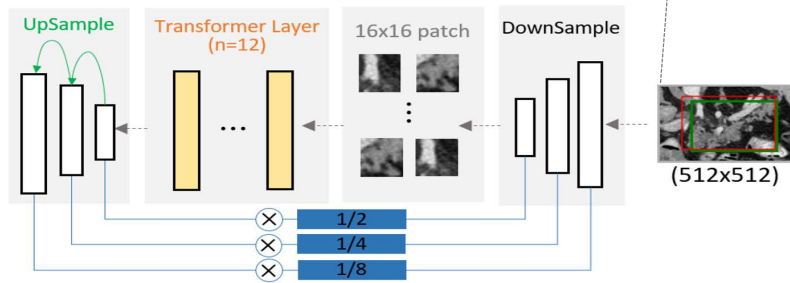


Pancreas
Yes / No



2-Stage

TransUnet(based Resnet50)



Evaluation

Loss Function

- Dice similarity Coefficient
$$\frac{2(A \cap B)}{A + B}$$

Where A and B are surfaces
- Dice Hausdorff distance
A: prediction
B: correct ones
$$\max_{s, a \in A} \min_{b \in B} \|a - b\|$$

Evaluation Metrics

- Max DSC
- Min DSC
- Mean DSC
- Precision
- Recall